



novatooth



COMPUTER SCIENCE
UC SANTA BARBARA

Special thanks to:



Renato Untalan

Giovanni Vigna

Chandra Krintz

Ankita Singh

Arnie Cooper

2016 - 2017

Meet the Team



Kevin Chan

kevinchan@umail.ucsb.edu



Albert Chen

albert_chen@umail.ucsb.edu



Trevor Morris

trevormorris@umail.ucsb.edu



Robert Stosick

rstosick@umail.ucsb.edu



Henry Yu

henryyu@umail.ucsb.edu



hello!

Introduction and Goal

- Penetration testing is a growing industry from rising security needs
- We want to find the exploits before the bad guys do
- Our platform allows penetration testers to achieve this

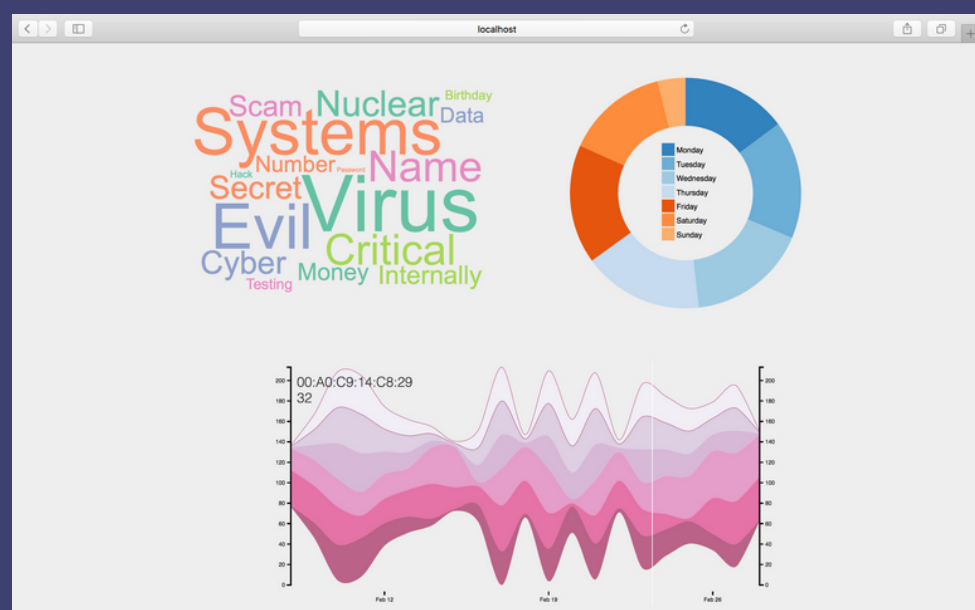
Development Workflow



Technologies Used



Spotlight: Data Visualization



Eavesdrop

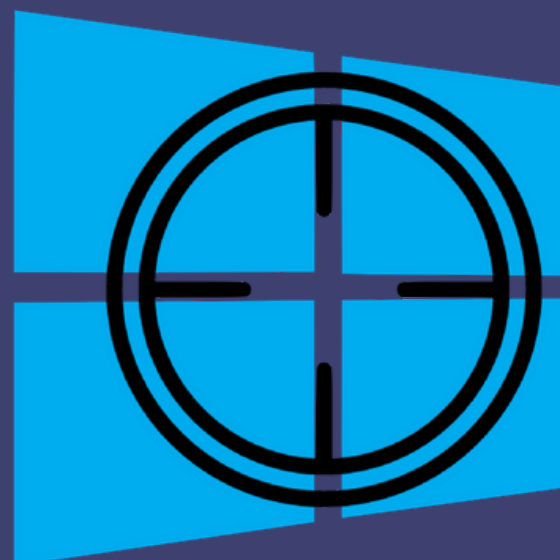
- A passive MitM attack on a bluetooth headset and a mobile phone. Right from the point of contact, it enables the user to listen in on all outgoing audio traffic from the phone and send that data to our web back-end for analytics.



BluePhish

- An active Man-in-the-Middle attack on a bluetooth headset and a mobile phone. Using phishing schemes, the attack is able to gather targeted sensitive data.

We are a platform



WinRecon

- Bluetooth analytics and survey software that allows users to continually scan for nearby Bluetooth devices on Windows-based computers and report that information to the cloud for storage and analysis.

Spotlight: Man-In-The-Middle



Phone: "Alice"

Device: "Eve"



Bluetooth Headset: "Bob"



Web Backend



Google Cloud Speech API

Problem

- Security flaws in Bluetooth technology
- Exponentially increasing usage of Bluetooth
- Rise of cyber warfare and terrorism
- Insufficient knowledge and exposure of potential risks

Solution

- Perform penetration testing
- Identify security vulnerabilities
- Increase security consciousness & awareness
- Facilitate mass data analysis
- Show that current security measures are not sufficient
- Push Bluetooth makers to overlay more security features & protocols